BT-203 (GS)

B.Tech., I & II Semester Examination, December 2025

Grading System (GS)

BASIC MECHANICAL ENGINEERING

Time : Three Hours Maximum Marks : 70

Note : (i) Attempt any five questions. (ii) All questions carry equal marks. (iii) Original paper bilingual (Hindi + English) tha; yahan sirf English version diya gaya hai.

Coverage: Full paper (Q1–Q8)

1. (a) Define alloy steel. Explain the effect of alloying elements on steel properties. **7**

(b) Describe important phases and critical points in the Iron–Carbon diagram with the help of sketch. **7**

2. (a) Define lathe. Explain the main parts of a lathe machine with neat sketch. **7**

(b) List the different types of temperature measuring instruments. Also explain the working of optical pyrometer. **7**

3. (a) Derive Bernoulli's equation for incompressible fluids. Also discuss its physical significance. **7**

(b) Explain the construction and working principle of a reciprocating pump. **7**

4. (a) Define draught. Explain natural draught and its working. **7**

(b) Define the following statements of the second law of thermodynamics: (i) Clausius statement (ii) Kelvin–Planck statement. **7**

5. (a) Explain the Diesel cycle with P–V and T–S diagram. Also derive the efficiency of the Diesel cycle. **7**

(b) Explain the working of a four-stroke petrol engine with neat sketch. **7**

6. (a) Describe tensile testing and explain stress–strain diagram of ductile materials. **7**

(b) Define compressor. Explain the working principle of a reciprocating air compressor. **7**

7. (a) Define boiler. Explain the classification of boilers, also explain any one of them. **7**

(b) Explain the following: (i) Homogeneous and heterogeneous fluids (ii) Newton's law of viscosity. **7**

8. (source me ye question available nahi tha — original paper dekhein) **—**

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